

CoreHPC Access Primer

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This primer introduces new users to the CoreHPC (High Performance Computing) cluster. It links to additional documentation that include examples, and SGE-to-SLURM conversions.

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Prerequisites

- A UCSF Campus Account.
 - Collaborators must have a UCSF affiliated account.
- A Core HPC provisioned account.
- Be connected to UCSFwpa (WiFi), or VPN (remote).
 - In the case that UCSFwpa does not directly connect, please connect to VPN (a VPN connection will still work while using UCSFwpa).
- Have a personal device enrolled with MFA/DUO.

Additional Guides:

- **SLURM Examples:** [Examples page](#).
 - **SGE to SLURM**
 - [short translation guide of how certain SGE/AGE/OGE commands map to SLURM](#).
 - **Python virtual environments?** [Python environments using Miniforge3](#).
 - CoreHPC supports Miniforge3.
 - **Frequently Asked Questions.**
 - This is an ongoing document that will be populated with the most common questions.
 - **Technical and User Support**
 - Please fill out the following form: <https://tiny.ucsf.edu/Gethpc>
 - **Slack Channel**
 - We also [have this slack channel to discuss CoreHPC issues](#) and topics.
-  • **CoreHPC notes:** Cluster systems share the same home directory across machines **except user bastions**, as those have separate home directories for security.
- Login and compute nodes provide **temporary** shared scratch storage at `/scratch/user/<username>` or `/scratch/group/<group>` for staging and job data.
 - Software is provided via OpenHPC modules (`module avail`, `module load <name>`) and is consistent across login and compute nodes. Please see the SLURM [Submission Examples page](#)

How to Connect to CoreHPC (Login Node)

 The **chpc-ucsf-login-vm1** is designed for submitting jobs into the cluster and not for prototyping or running programs. Please be mindful to not attempt running heavy computational programs in it.

  **Note:** Standard CoreHPC bastions do not allow `scp` or `rsync` for data transfers, per security policy. All data transfers will soon be required to use **Globus**. As a temporary workaround, early access users may use the **administrative bastion** for `scp/rsync`; this access will be removed once Globus is available.

1. Connect to the UCSF VPN.
2. Use an SSH client to connect to one of the user access bastion nodes:

```
ssh <campus username>@chpc-ucsf-bastion-vm1.corehpc.ucsf.edu
```

- a. Alternate bastion: `chpc-ucsf-bastion-vm2.corehpc.ucsf.edu`
3. Respond to the **MFA/DUO** verification challenge.
4. From the user access bastion use `ssh` to connect to the CoreHPC login node. This is the node that you will launch your SLURM cluster jobs from.

```
[myself@chpc-ucsf-bastion-vm1 ~] ssh chpc-ucsf-login-vm1
```



Please note that

How To List Available Modules (e.g. module avail)

An Example of CoreHPC Available Modules

[Expand source](#)

For the **latest and greatest** list of modules, be sure to run "**module avail**" when logged in chpc-ucsf-login-vm1.

How To Transfer Data to CoreHPC

 Please be aware that **temporary** means that these scratch areas will be subject to a periodic cleaning policy. The storage is meant to be **temporary**, reach out to us if a particular project requires a specific amount of storage time for a given dataset to discuss service options. Also, there is a quota of 20G for home directories and a maximum user quota of 50TB for the scratch directory.

CoreHPC offers a **temporary** high speed storage location under /mnt/scratch/\$USER, where \$USER is your personal scratch area. We can also create a group location for **temporary** storage with the same name as your CoreHPC-mounted FAC share (e.g. CX000000_DS0). Please see the FAQ if you want to know how to acquire a Globus.

There are two methods of moving data to and from CoreHPC.

- CoreHPC mounted FAC share (you must have an existing FAC share, or request one from the FAC service: <https://tiny.ucsf.edu/getfac>).
- Globus transfer service.

FAC Shares

For users with CoreHPC mounted FAC shares, you can use commands like cp or rsync in the login node:

```
#cp -R /mnt/fac/<your FAC share name>/[file|dir] /mnt/scratch/group/<your FAC share name>/
#rsync -av /mnt/fac/<your FAC share name>/[file|dir] /mnt/scratch/group/<your FAC share name>/
#
#Example
cp -R /mnt/fac/CX000123_DS1/MyFolder /mnt/scratch/group/CX000123_DS1/
```

Globus Service

1. Log into the Globus site: <https://www.globus.org/> using your UCSF Globus account.
2. Select the first item on the left side "File Manager".
3. In the "Search" field, type "corehpc".
4. Select a collection.
 - a. UCSF CoreHPC - User Home
 - b. UCSF CoreHPC - User Scratch
5. Look for and select your directory, conveniently named after your UCSF Campus login name.
6. Use any of the file operations (e.g. upload) from the panel on the right side.

The files will be accessible in the **login node** in our respective /home/remote/<username> or /mnt/scratch/user/<username>

**UCSF CoreHPC - User Home**Subscribed Mapped Collection (GCS) on **CoreHPC UCSF Endpoint 1**

Owner: ucsfcorehpc@globusid.org

Domain: m-5b3ad74b6728.854f.gaccess.io

**UCSF CoreHPC - User Scratch**Subscribed Mapped Collection (GCS) on **CoreHPC UCSF Endpoint 1**

Owner: ucsfcorehpc@globusid.org

Domain: m-03670c.4b6728.854f.gaccess.io

Technical Issues and User Support

For any issues related to CoreHPC, including technical and user support, please fill out and submit the following form:
<https://tiny.ucsf.edu/Gethpc>

Basic SLURM Workflow

The following procedure outlines a typical SLURM workflow, please review the additional guides for further examples.

1. List available resources in SLURM.
2. Submit a job.
3. Verify the status of the job.
4. Allow it to complete.
5. If for any reason the job has stalled, cancel it and try again.

List Available Resources.

```
sinfo -o "%n %e %m %a %c %C %G"
```

Submit your script with sbatch.

```
$ sbatch <OPTIONS> example_batch_script.sh
```

```
example_batch_script.sh
```

[Expand source](#)

Check the status of your job.

```
$ squeue
```

```
squeue example
```

[Expand source](#)

Basic Workflow Task Commands

Examples on how to use these can be found at the CoreHPC [Submission Examples page](#).

Task	Command
Submit a batch job	<code>sbatch job.sh</code>
Run an interactive job	<code>salloc</code>
Run a command with SLURM	<code>srun <command></code>
View job queue	<code>squeue</code>
View only your jobs	<code>squeue -u \$USER</code>
Cancel a job	<code>scancel <jobid></code>
View job details	<code>scontrol show job <jobid></code>
Check resource usage	<code>sacct</code>

Note: These commands can also be invoked with "<command> --help" to list their usage parameters, or "man <command>" to display the Linux manual page for the command.

Job Status Codes

idle	Node is ready to accept jobs.
mix	Node has running jobs/resources allocated.
down	Soft down. Possible network communication, or SLURMD stopped.
down*	Hard down. Slurmd offline, and node possibly powered off.
maint	Node is down for scheduled maintenance.
drain	The node is unavailable for use due to maintenance or errors.
plnd	Marked for maintenance.
comp	The job is completing.
drng	The node is in the process of draining.
resv	A reservation has been set on the node.

Cluster Resource Capabilities

CPU

No labels

Need help? Check out the [Wiki Help](#) space or the [FAQs](#) for answers to common issues.

